Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/24/2018
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	djdurian@physics.upenn.edu	**Fax:**	FY 215-898-2010
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-8147
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	209 South 33rd Street		
PI Web Page:	http://www.physics.upenn.edu/		
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	07/21/2014	**End Date:**	07/20/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Tin, Padetha	**Contact Phone:**	216-433-8164
Contact Email:	Padetha.Tin@grc.nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AM99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE (May 2015): Continuation of "The Melting of Aqueous Foams"; previous grant numbers NNX07AP20G ; NNX07AK60G ; NNC04GB61G. NOTE (4/22/2019): Continued by "Foam Optics and Mechanics," grant 80NSSC19K0599 with the same Principal Investigator, Douglas Durian, Ph.D. The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into Advanced Life Support (ALS) technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ALS, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this cycle my FOAM group consisted of one PhD student, Anthony Chieco and one undergraduate, Cody Schimming. The focus of our work was on ground-based studies of the structure and time evolution of coarsening foams, in support of the planned flight experiments with experiment container SOFT MATTER DYNAMICS (formerly FOAM-C). This program is being financed and directed by the European Space Agency (ESA). The instrument itself has already been built by the contractor, Astrium. The measurement campaign aboard the International Space Station (ISS) is scheduled for early 2019. The first round of sample cells will be devoted to the coarsening of foams, the fluid/glassy dynamics of granular matter, and fundamentals of light scattering. The FOAM focus is on structure, stability, and temporal evolution, for sequences of progressively wetter foams. These cannot be studied on Earth due to gravitation drainage, which increases with wetness and effectively prevents study of behaving when the bubbles become nearly spherical and unjammed. At Penn, we have made progress in several areas during the current funding period. This includes theoretical work on how gas is transported across Plateau borders in wet foams. In the past, borders have been assumed to totally block the gas; we show that much more is transported than naïve scaling expectations. Work also includes the introduction of a new way to search for hidden order in amorphous systems, using a new concept for how to diagnose hyperuniformity in terms of the length scale over which fluctuations are important. These topics led to four peer-reviewed publications, listed in the below Bibliography section. Important extensions are now in progress and will be used in analyzing data from the flight experiments. MANUSCRIPTS UNDER REVIEW: A. T. Chieco, M. Zu, A. J. Liu, N. Xu and D. J. Durian, Uniformity of Soft Disk Packings above and Below Jamming, submitted to Phys. Rev. Lett., (2018). S. Aumaître, R. Behringer, A. Cazaubiel, E. Clément, J. Crassous, D. Durian, E. Falcon, S. Fauve, D. Fischer, A. Garcimartin, Y. Garrabos, X. J. Meiying Hou, C. Lecoutre, S. Luding, D. Maza, M. Noirhomme, E. Opsomer, F. Palencia, T. Pöschel, J. Schockmel, M. Sperl, R. Stannarius, N. Vandewalle and P. Yu, An Instrument for Studying Granular Media in Low-Gravity Environment, submitted to Rev. Sci. Inst., (2018). [Ed. note 4/22/19: now published and see Bibliography section] ED. NOTE (4/22/2019): Continued by "Foam Optics and Mechanics," grant 80NSSC19K0599 with the same Principal Investigator, Douglas Durian, Ph.D.

Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Chieco AT, Schimming CD, Durian DJ. "Experimentally Testing the Border-Crossing Model for Coarsening of Quasi-Two-Dimensional Wet Foams." Presented at Annual March Meeting of the American Physical Society, Los Angeles, CA, March 5-9, 2018. Bulletin of the American Physical Society. 2018 Mar;3(4):S54.00004. http://meetings.aps.org/Meeting/MAR18/Session/S54.4 ; accessed 4/24/18. , Mar-2018
Articles in Peer-reviewed Journals	Aumaître S, Behringer R, Cazaubiel A, Clément E, Crassous J, Durian D, Falcon E, Fauve S, Fischer D, Garcimartin A, Garrabos Y, Meiying Hou XJ, Lecoutre C, Luding S, Maza D, Noirhomme M, Opsomer E, Palencia F, Pöschel T, Schockmel J, Sperl M, Stannarius R, Vandewalle N, Yu P. "An instrument for studying granular media in low-gravity environment." Review of Scientific Instruments. 2018 July;89(7):075103. https://doi.org/10.1063/1.5034061 ; PubMed PMID: 30068123 , Jul-2018
Articles in Peer-reviewed Journals	Zayed A, Badruddoza M, MacWilliams SV, Sebben DA, Krasowska M, Durian DJ, Ferri JK. "Diffusing wave spectroscopy (DWS) methods applied to double emulsions." Current Opinion in Colloid & Interface Science. 2018 Sep;37:74-87. https://doi.org/10.1016/j.cocis.2018.06.006 , Sep-2018
Articles in Peer-reviewed Journals	Chieco AT, Zu M, Liu AJ, Xu N, Durian DJ. "Spectrum of structure for jammed and unjammed soft disks." Physical Review E. 2018 Oct;98(4):042606. https://doi.org/10.1103/PhysRevE.98.042606 , Oct-2018
Articles in Peer-reviewed Journals	Schimming CD, Durian DJ. "Border-crossing model for the diffusive coarsening of two-dimensional and quasi-two-dimensional wet foams." Physical Review E. 2017 Sep;96(3-1):032805. https://doi.org/10.1103/PhysRevE.96.032805 ; PubMed PMID: 29346872 , Sep-2017
Articles in Peer-reviewed Journals	Chieco AT, Dreyfus R, Durian DJ. "Characterizing pixel and point patterns with a hyperuniformity disorder length." Physical Review E. 2017 Sep;96(3-1):032909. https://doi.org/10.1103/PhysRevE.96.032909 ; PubMed PMID: 29346987 , Sep-2017
Articles in Peer-reviewed Journals	Durian DJ. "Hyperuniformity disorder length spectroscopy for extended particles." Physical Review E. 2017 Sep;96(3-1):032910. https://doi.org/10.1103/PhysRevE.96.032910 ; PubMed PMID: 29346949 , Sep-2017